

PROBLEM 1

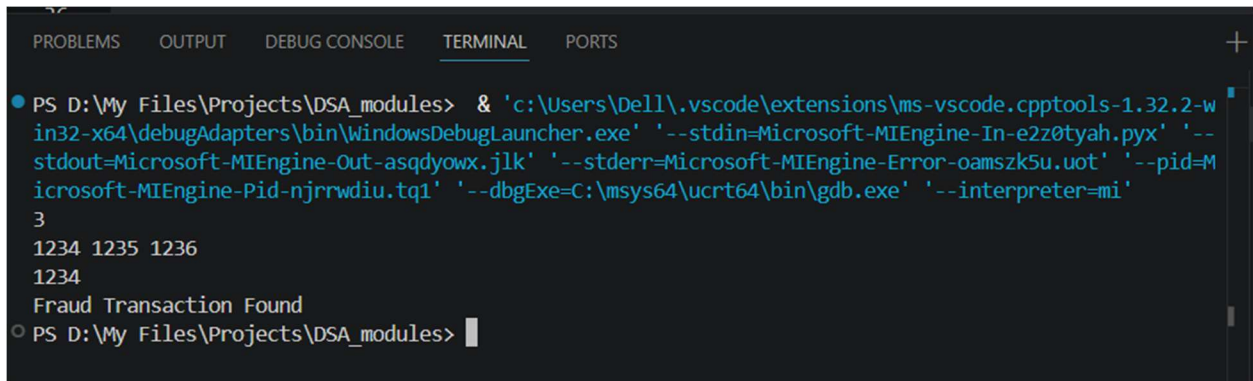
CODE:

```
module2 > linear_search.cpp > ...
3  #include <iostream>
4  using namespace std;
5
6  int main()
7  {
8      int n, key;
9      cin >> n;
10
11     int arr[n];
12
13     for(int i = 0; i < n; i++)
14     {
15         cin >> arr[i];
16     }
17
18     cin >> key;
19
20     bool found = false;
21
22     for(int i = 0; i < n; i++)
23     {
24         if(arr[i] == key)
25         {
26             found = true;
27             break;
28         }
29     }
30
31     if(found)
32     {
33         cout << "Fraud Transaction Found";
34     }
35     else
36     {
37         cout << "Transaction Not Found";
38     }
39
40     return 0;
41 }
```

LOGIC:

```
/*
Logic:
1. Read number of transaction IDs.
2. Store all IDs in an array.
3. Read the suspicious transaction ID.
4. Traverse the array one by one.
5. If the ID matches, mark it as found and stop searching.
6. Print "Fraud Transaction Found" if found, otherwise print "Transaction Not Found".
*/
```

OUTPUT:

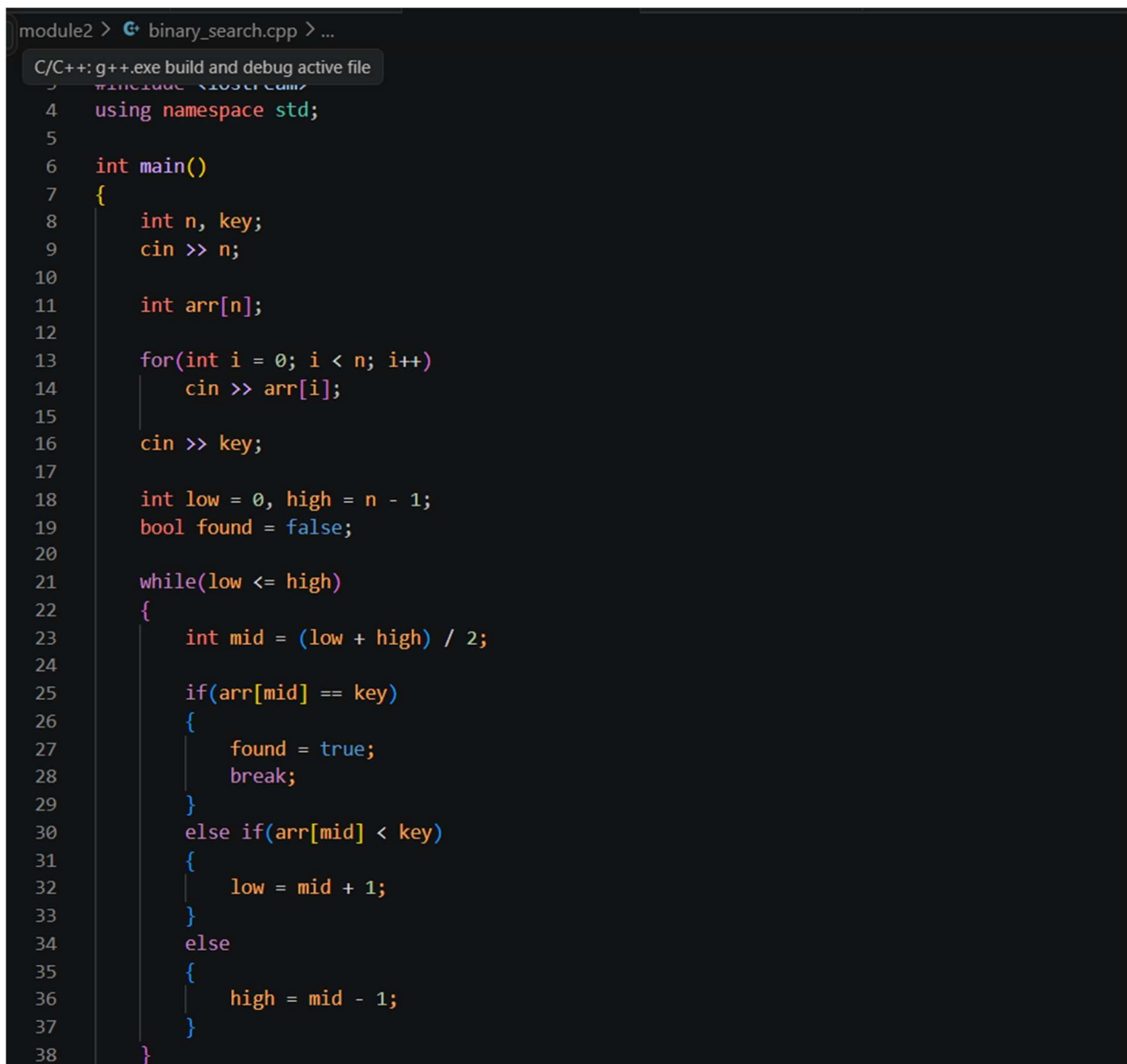


A screenshot of a Visual Studio Code terminal window. The terminal shows a command prompt where a C++ program is being executed with various debug options. The output of the program is displayed below the command line.

```
PS D:\My Files\Projects\DSA_modules> & 'c:\Users\Dell\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-e2z0tyah.pyx' '--stdout=Microsoft-MIEngine-Out-asqdyowx.jlk' '--stderr=Microsoft-MIEngine-Error-oamszk5u.uot' '--pid=Microsoft-MIEngine-Pid-njrrwdui.tq1' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
3
1234 1235 1236
1234
Fraud Transaction Found
PS D:\My Files\Projects\DSA_modules>
```

PROBLEM 2

CODE:



A screenshot of a Visual Studio Code editor window showing a C++ program for binary search. The code is written in a dark theme with syntax highlighting. The program includes a main function that reads an array and a key, then performs a binary search to find the key in the array.

```
module2 > binary_search.cpp > ...
C/C++: g++.exe build and debug active file
1  #include <iostream>
2
3
4  using namespace std;
5
6  int main()
7  {
8      int n, key;
9      cin >> n;
10
11     int arr[n];
12
13     for(int i = 0; i < n; i++)
14     {
15         cin >> arr[i];
16     }
17
18     cin >> key;
19
20     int low = 0, high = n - 1;
21     bool found = false;
22
23     while(low <= high)
24     {
25         int mid = (low + high) / 2;
26
27         if(arr[mid] == key)
28         {
29             found = true;
30             break;
31         }
32         else if(arr[mid] < key)
33         {
34             low = mid + 1;
35         }
36         else
37         {
38             high = mid - 1;
39         }
40     }
41 }
```

```

29         }
30         else if(arr[mid] < key)
31         {
32             low = mid + 1;
33         }
34         else
35         {
36             high = mid - 1;
37         }
38     }
39
40     if(found)
41         cout << "Log Found";
42     else
43         cout << "Log Not Found";
44
45     return 0;
46 }

```

LOGIC:

```

/*
Logic:
1. Read the sorted array.
2. Set low = 0 and high = n-1.
3. Find the middle element.
4. If middle element equals target, search ends.
5. If target is greater, search right half.
6. If target is smaller, search left half.
7. Continue until found or low becomes greater than high.

*/

```

OUTPUT:

```

PS D:\My Files\Projects\DSA_modules> & 'c:\Users\Dell\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-cw5nwhbc.1ht' '--stdout=Microsoft-MIEngine-Out-amgnzyyz.ouf' '--stderr=Microsoft-MIEngine-Error-xypfd14y.1bs' '--pid=Microsoft-MIEngine-Pid-gdaugbps.wgm' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
3
1234 1235 1236
1234
Log Found
PS D:\My Files\Projects\DSA_modules>

```

PROBLEM 3

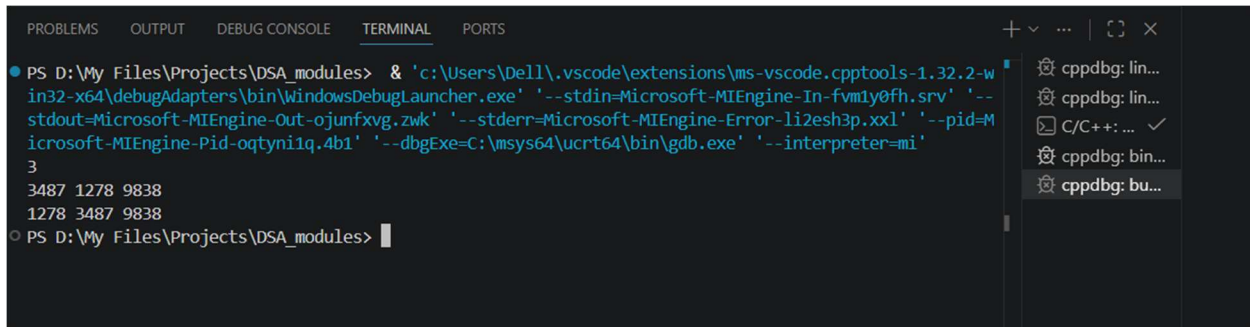
CODE:

```
module2 > bubble_sort.cpp > ...
1  /* 3. Sort Employee Salaries (Bubble Sort) */
2
3
4  #include <iostream>
5  using namespace std;
6
7  int main()
8  {
9      int n;
10     cin >> n;
11
12     int arr[n];
13
14     for(int i = 0; i < n; i++)
15         cin >> arr[i];
16
17     for(int i = 0; i < n - 1; i++)
18     {
19         for(int j = 0; j < n - i - 1; j++)
20         {
21             if(arr[j] > arr[j + 1])
22             {
23                 int temp = arr[j];
24                 arr[j] = arr[j + 1];
25                 arr[j + 1] = temp;
26             }
27         }
28     }
29
30     for(int i = 0; i < n; i++)
31         cout << arr[i] << " ";
32
33     return 0;
34 }
35
```

LOGIC:

```
35
36  /*
37  Logic:
38  1. Read all salaries into an array.
39  2. Compare adjacent elements.
40  3. Swap them if they are in the wrong order.
41  4. After every pass, the largest element reaches the end.
42  5. Repeat until the array is sorted.
43  6. Print the sorted array.
44
45
46  */
47
```

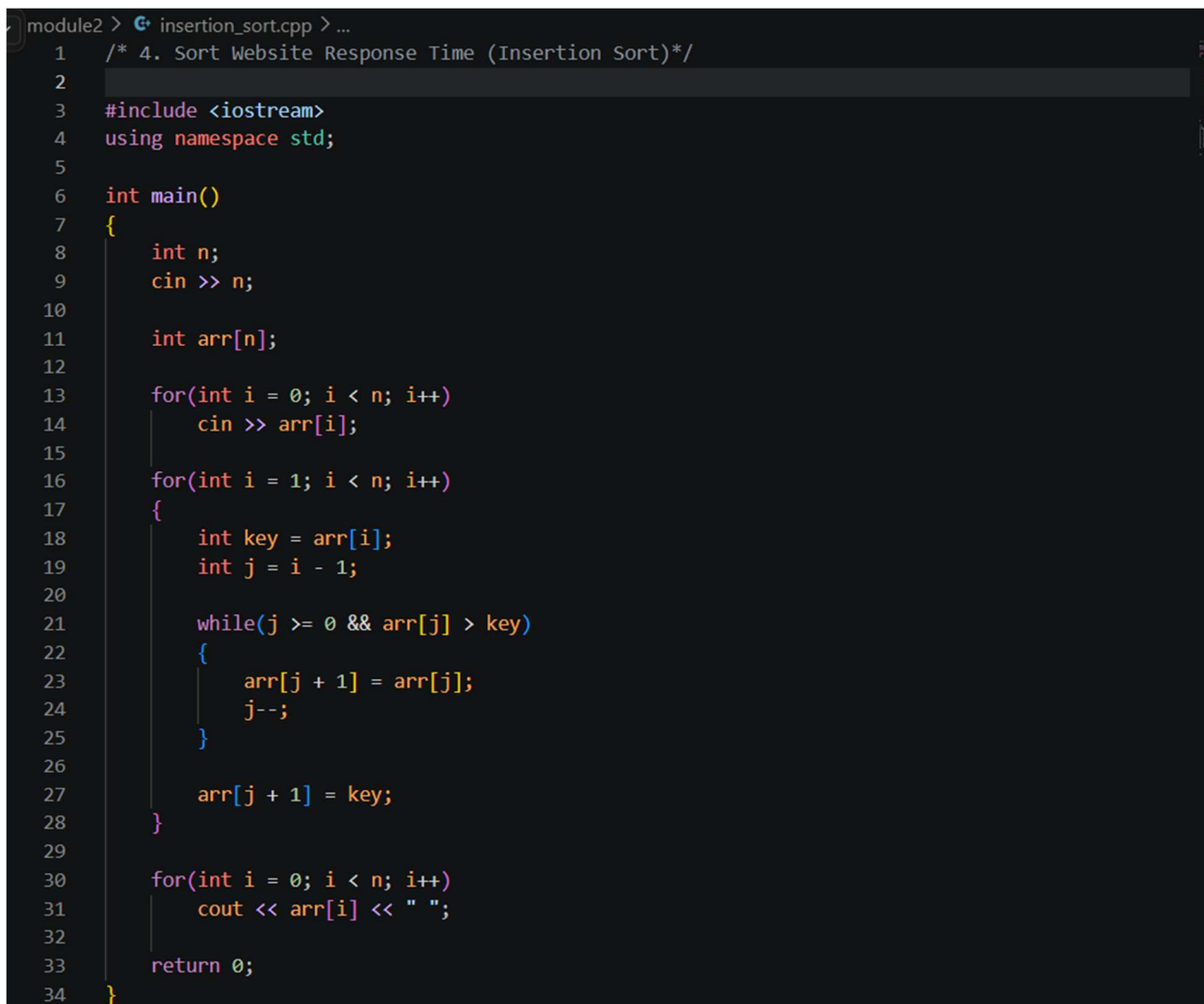
OUTPUT:



```
PS D:\My Files\Projects\DSA_modules> & 'c:\Users\Dell\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-fvm1y0fh.srv' '--stdout=Microsoft-MIEngine-Out-ojunfxvg.zwk' '--stderr=Microsoft-MIEngine-Error-li2esh3p.xml' '--pid=Microsoft-MIEngine-Pid-oqtyni1q.4b1' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
3
3487 1278 9838
1278 3487 9838
PS D:\My Files\Projects\DSA_modules>
```

PROBLEM 4

CODE:



```
module2 > insertion_sort.cpp > ...
1  /* 4. Sort Website Response Time (Insertion Sort)*/
2
3  #include <iostream>
4  using namespace std;
5
6  int main()
7  {
8      int n;
9      cin >> n;
10
11     int arr[n];
12
13     for(int i = 0; i < n; i++)
14         cin >> arr[i];
15
16     for(int i = 1; i < n; i++)
17     {
18         int key = arr[i];
19         int j = i - 1;
20
21         while(j >= 0 && arr[j] > key)
22         {
23             arr[j + 1] = arr[j];
24             j--;
25         }
26
27         arr[j + 1] = key;
28     }
29
30     for(int i = 0; i < n; i++)
31         cout << arr[i] << " ";
32
33     return 0;
34 }
```

LOGIC:

```

/*
Logic:
1. Read all response times into an array.
2. Assume the first element is already sorted.
3. Pick the next element (key).
4. Compare the key with previous elements.
5. Shift all larger elements one position to the right.
6. Insert the key into its correct position.
7. Repeat until the entire array is sorted.
8. Print the sorted array.

*/

```

OUTPUT:

```

PS D:\My Files\Projects\DSA_modules> & 'c:\Users\Dell\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-vztarphw.ec2' '--stdout=Microsoft-MIEngine-Out-a4hm42yo.olo' '--stderr=Microsoft-MIEngine-Error-qxlco0bo.saw' '--pid=Microsoft-MIEngine-Pid-cijpcarx.lso' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
4
23 54 13 39
13 23 39 54
PS D:\My Files\Projects\DSA_modules>

```

PROBLEM 5

CODE:

```
module2 > selection_sort.cpp > ...
1  /*5. Prioritize Bug Reports (Selection Sort) */
2
3
4  #include <iostream>
5  using namespace std;
6
7  int main()
8  {
9      int n;
10     cin >> n;
11
12     int arr[n];
13
14     for(int i = 0; i < n; i++)
15         cin >> arr[i];
16
17     for(int i = 0; i < n - 1; i++)
18     {
19         int minIndex = i;
20
21         for(int j = i + 1; j < n; j++)
22         {
23             if(arr[j] < arr[minIndex])
24                 minIndex = j;
25         }
26
27         int temp = arr[i];
28         arr[i] = arr[minIndex];
29         arr[minIndex] = temp;
30     }
31
32     for(int i = 0; i < n; i++)
33         cout << arr[i] << " ";
34
35     return 0;
36 }
```

LOGIC:

```
/*
Logic:
1. Read all priority scores into an array.
2. Assume the first unsorted element is the minimum.
3. Search the remaining array for a smaller element.
4. If found, update the minimum index.
5. Swap the smallest element with the first unsorted element.
6. Move the sorted boundary one step forward.
7. Repeat until the entire array is sorted.
8. Print the sorted array.

*/
```

OUTPUT:

```
42 3. Search the remaining array for a smaller element.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\My Files\Projects\DSA_modules> & 'c:\Users\Dell\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\windowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-z5bhohzhz.phg' '--stdout=Microsoft-MIEngine-Out-pp30asva.orm' '--stderr=Microsoft-MIEngine-Error-ghqjx2f4.div' '--pid=Microsoft-MIEngine-Pid-x1qmnza5.slx' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
5
9 4 2 6 0
0 2 4 6 9
PS D:\My Files\Projects\DSA_modules>
```

cppdbg: lin...
cppdbg: lin...
C/C++: ... ✓
cppdbg: bin...
cppdbg: bu...
cppdbg: ins...
cppdbg: ins...
cppdbg: sel...